TIC 378839425

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        0.757361 d
  Epoch (t0)    1569.8230 BTJD
  Duration      0.5000 d
  Depth         0.816 %  (8,163 ppm)
  SNR           214.1

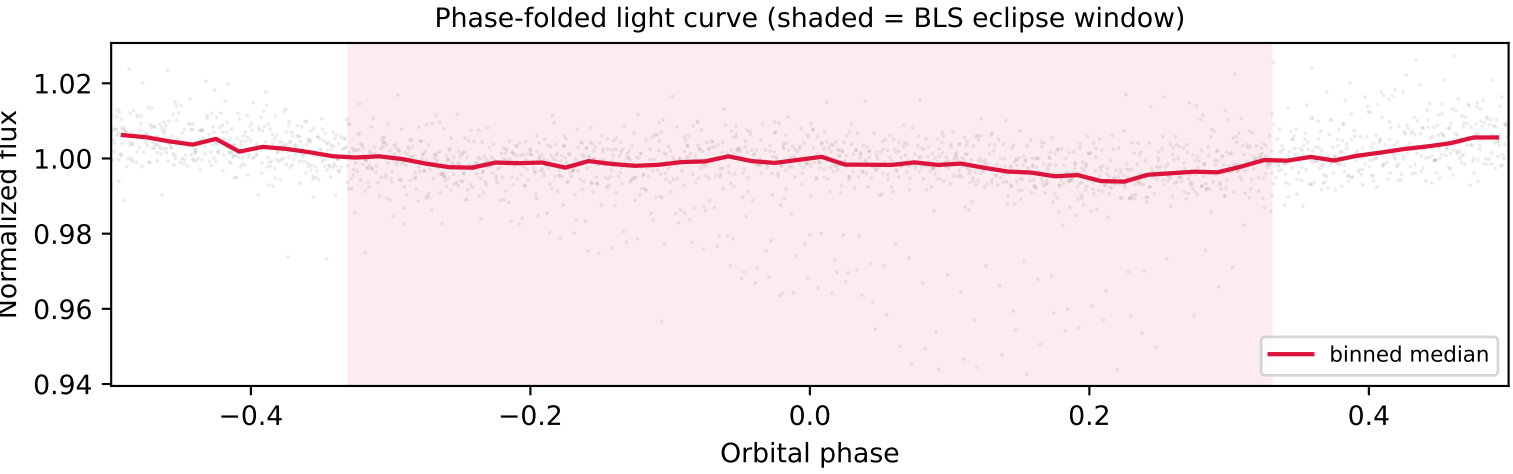
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      70  (gate >= 3)
  Robust depth ratio     0.60  (gate >= 0.50)
  Robust depth sigma     26.0  (gate >= 7.0)
  Odd/even z-score       1.46  (consistent)
  Secondary eclipse      not detected
```

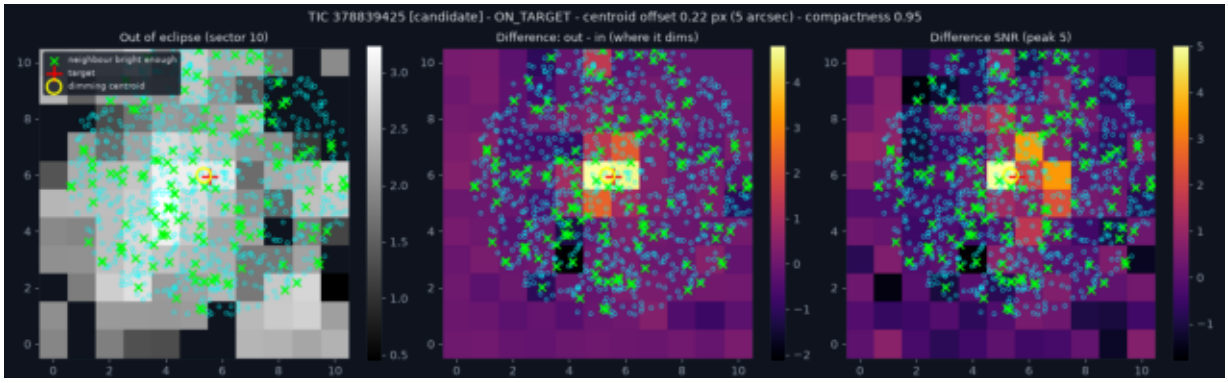
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 14.7 sigma    t0 refined by 0.2275 phase
  Difference-image peak SNR   5.01   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.22 px (5 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 378839421
    0.41 px away, dTmag +4.43 - farther from the dimming than the target
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No caveats - clean detection on every check.